



ASEAN
DATA SCIENCE
EXPLORERS

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Country: Vietnam

SEEK WELL

Democratizing Early Skin Cancer Detection
Through AI and Community Workers

SDGs to achieve:

3 GOOD HEALTH
AND WELL-BEING



11 SUSTAINABLE CITIES
AND COMMUNITIES



13 CLIMATE
ACTION



A skin cancer crisis in plain sight



Friday, November 15, 2024, 13:55 (GMT+7)

Skin cancer detected from black spots that appeared 10 years ago

HANOI: — A 70-year-old man had a black lesion on the sole of his foot for 10 years, thought to be a mole, but it gradually increased in size, and the results showed melanoma.



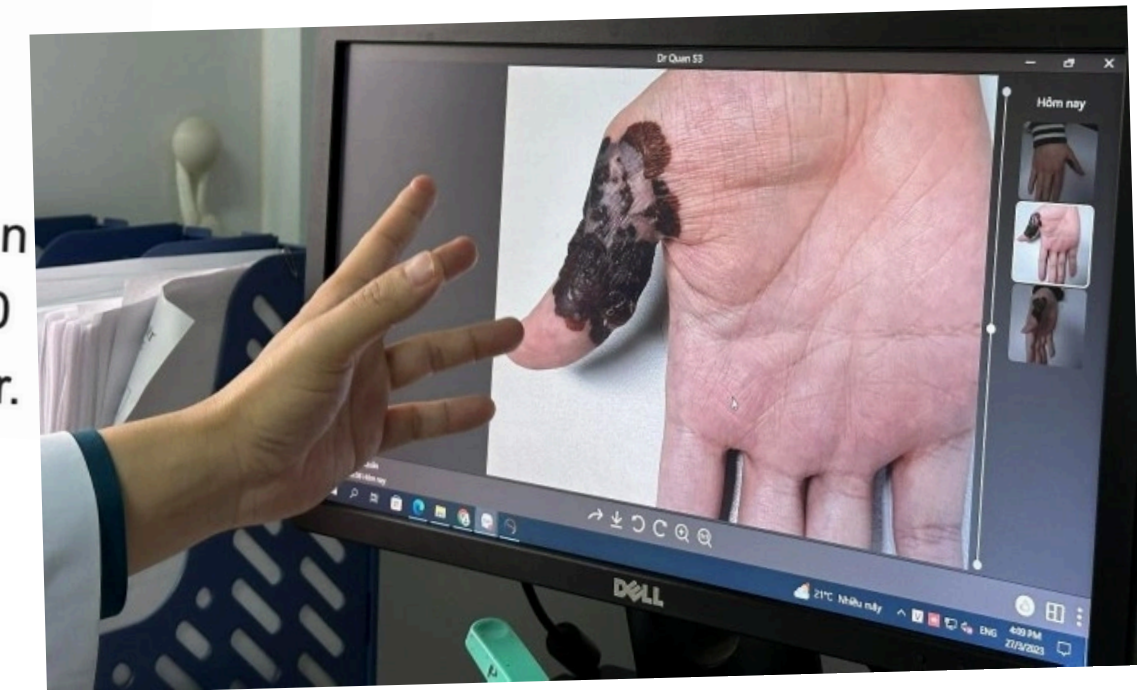
Source: vnexpress.net



Saturday, November 23, 2024, 05:00 (GMT+7)

The rate of Vietnamese people with skin cancer is increasing.

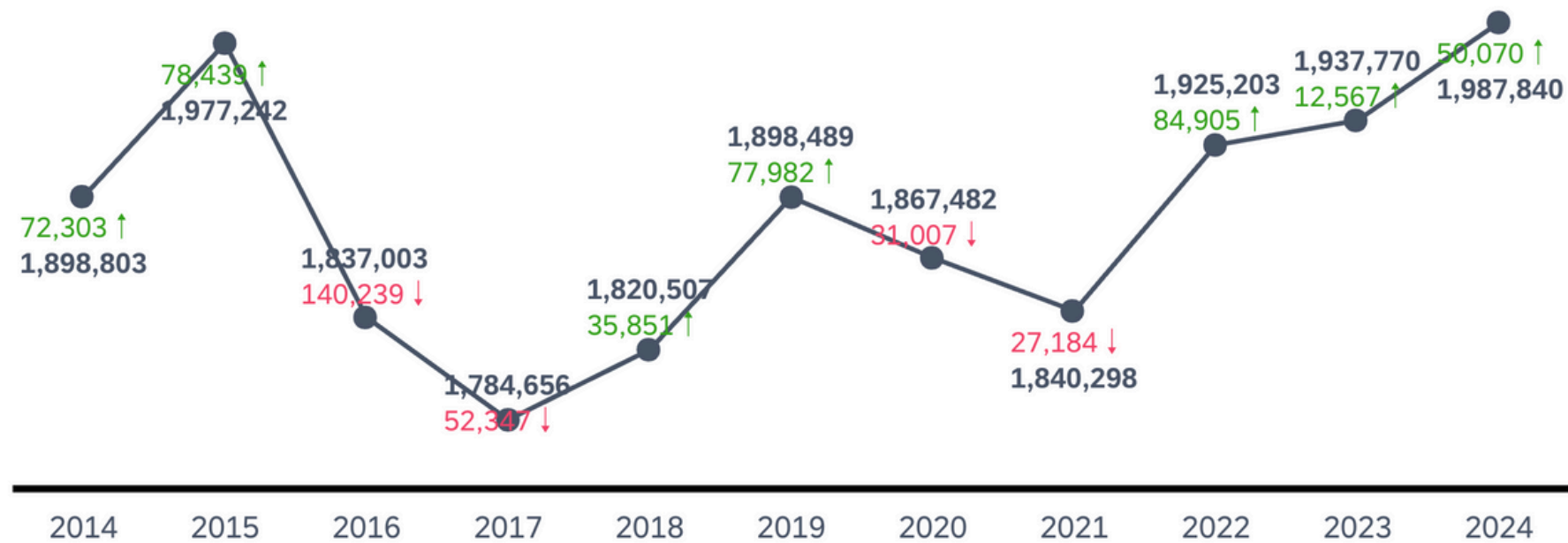
About 500 skin cancer patients come to the Central Dermatology Hospital for examination each year, a 50% increase compared to 5-10 years ago and with a tendency to be younger.



But what are the root causes driving this crisis?

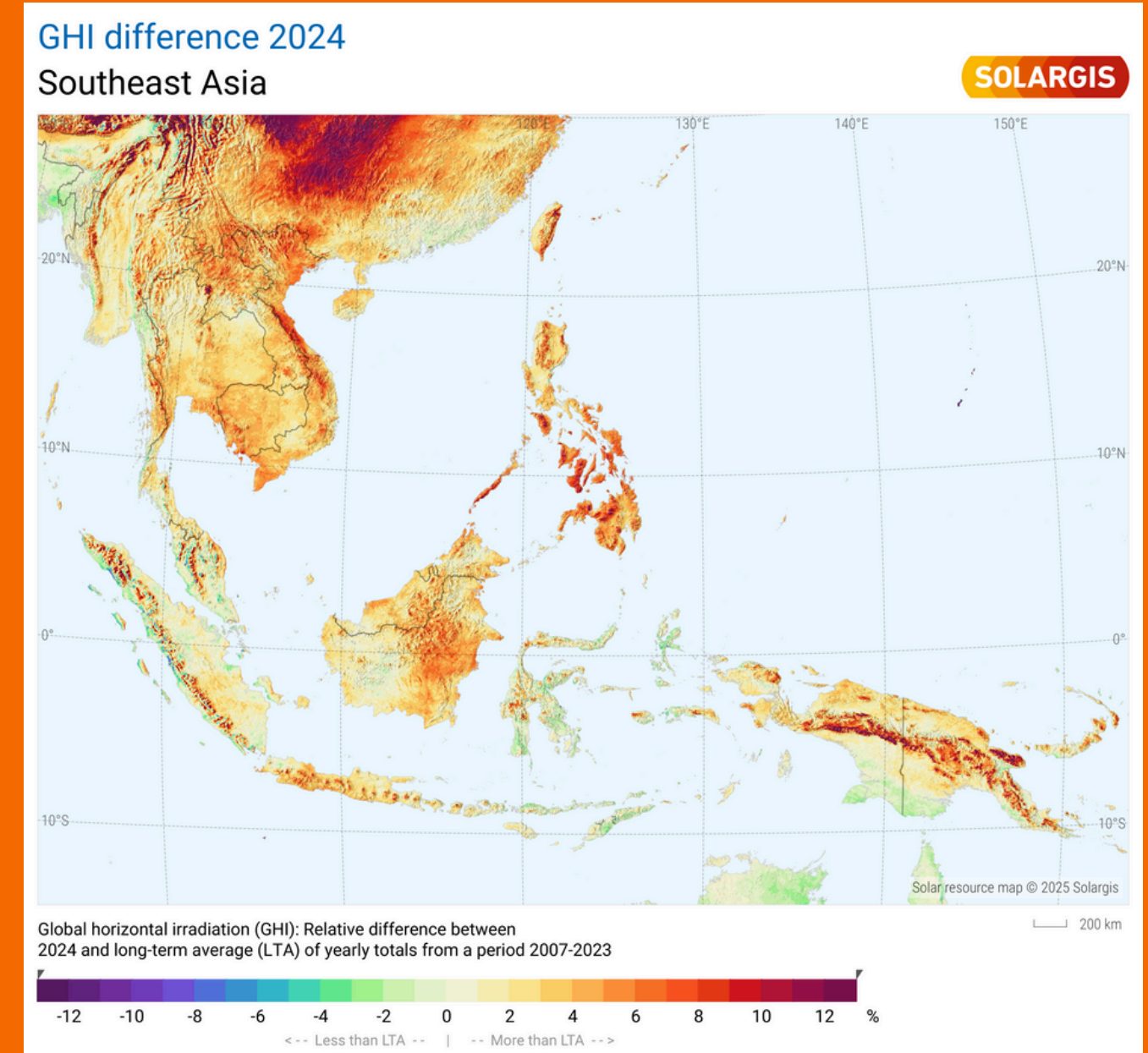
The Regional Driver – A Harsher Sun

Shortwave radiation (W/m²) per year, Ho Chi Minh city



In 2024, the solar radiation on Ho Chi Minh city has reached its highest in decades. And it is still increasing, not just in a single city.

Source: open-meteo.com



It's a regional crisis is fueled by **climate change**. In 2024, vast areas of Southeast Asia received up to 12% more solar radiation than the long-term average.

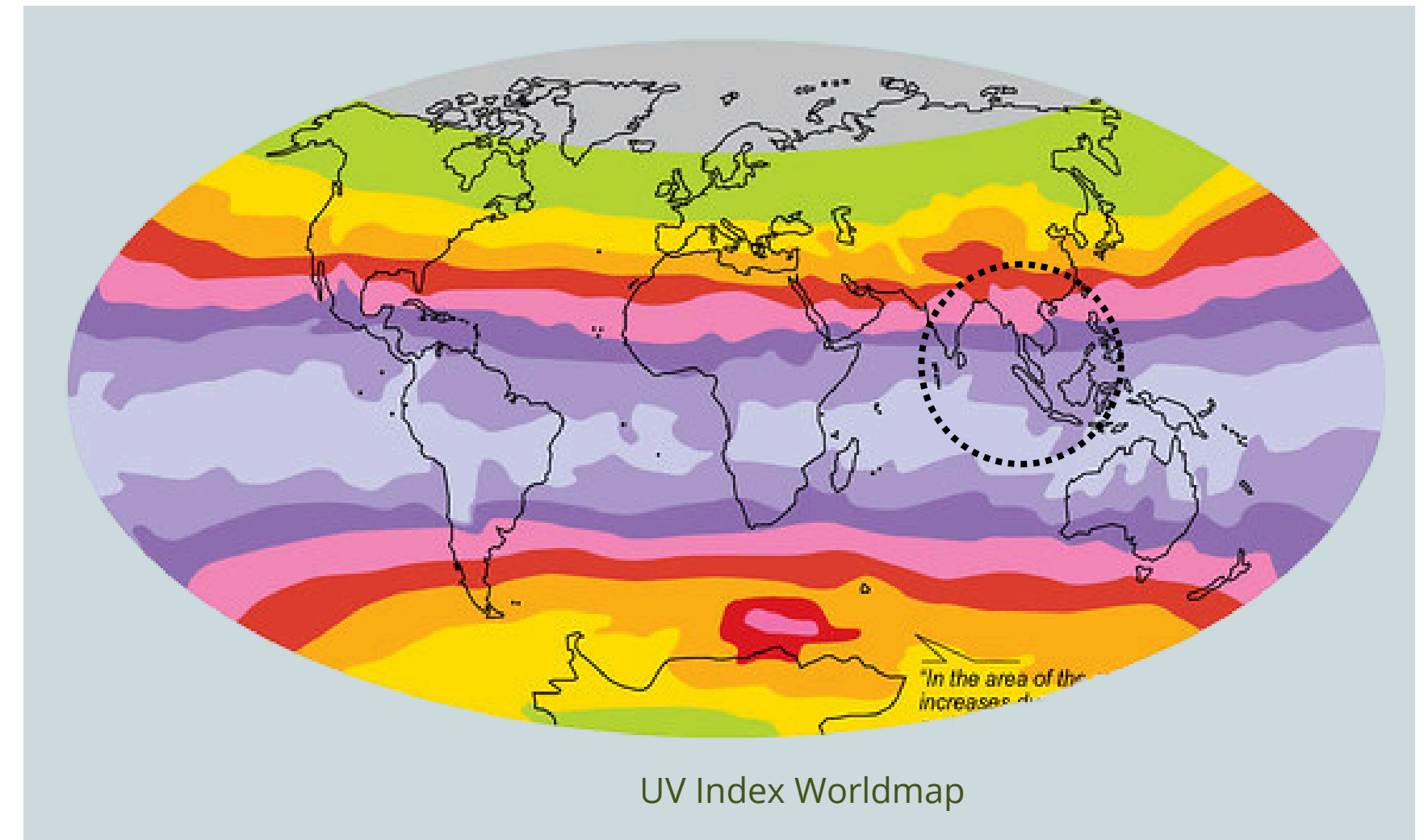
Source: solargis.com

Why is UV Radiation a Critical Threat in ASEAN?

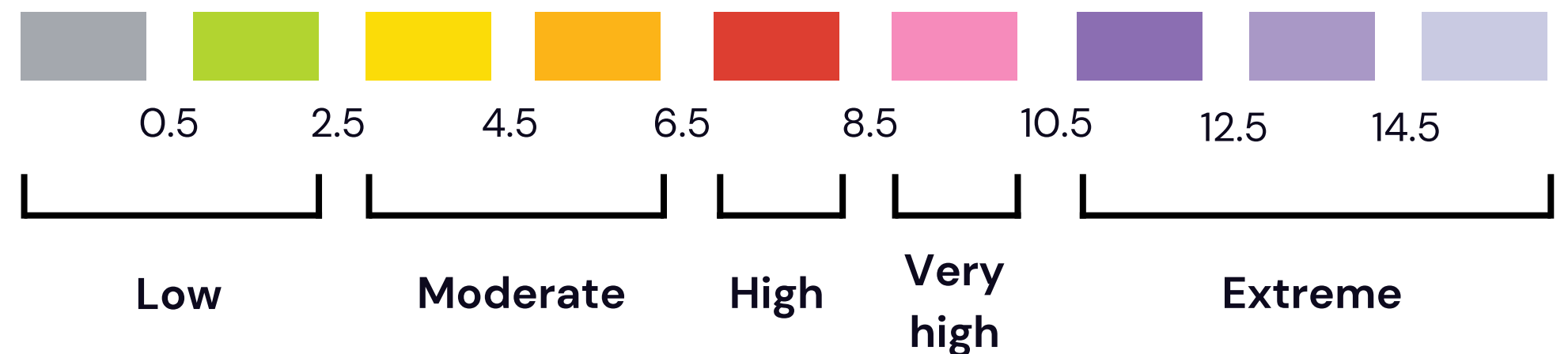
ASEAN – A permanent UV danger zone

Most ASEAN countries experience “very high” to “extreme” UV levels almost year-round.

Unlike many regions, there is no safe season – exposure risk remains constant.



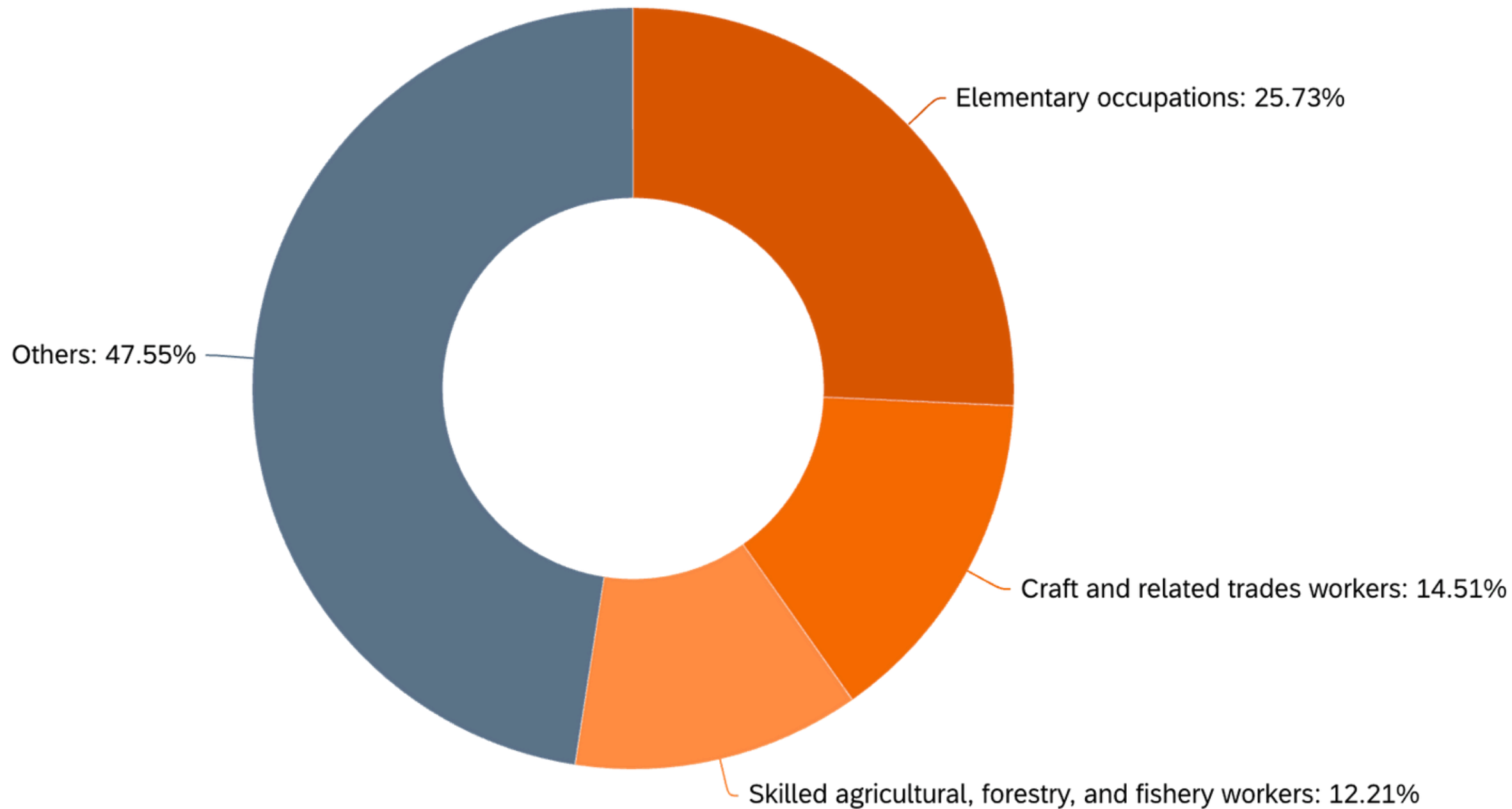
Daily maximum of the UV index by clear sky



Data source: UV Index WorldMap | GRID-Arendal. (n.d.).

The Unseen Frontline

– A Nation Working Outdoors



Over half the labor force in Vietnam (more than a third in ASEAN) are at risk of working outdoors, being exposed every day to solar radiation, including UV radiation

Source: General Statistics Office of Vietnam

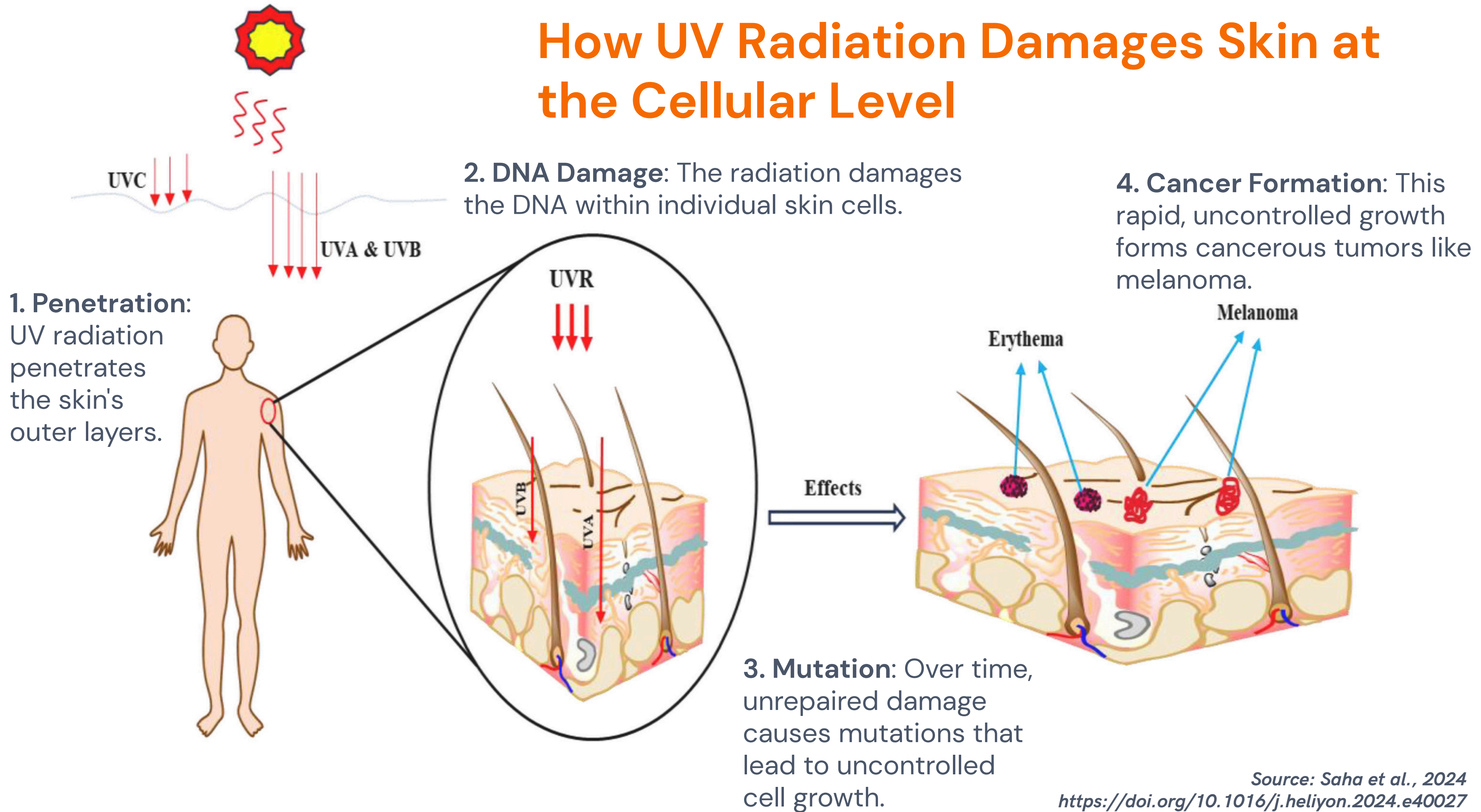
Over **80%** of skin cancers are caused by overexposure to UV radiation

More than **90%** Have experienced diseases related to extreme weather

But, only **27%** Seek for professional help

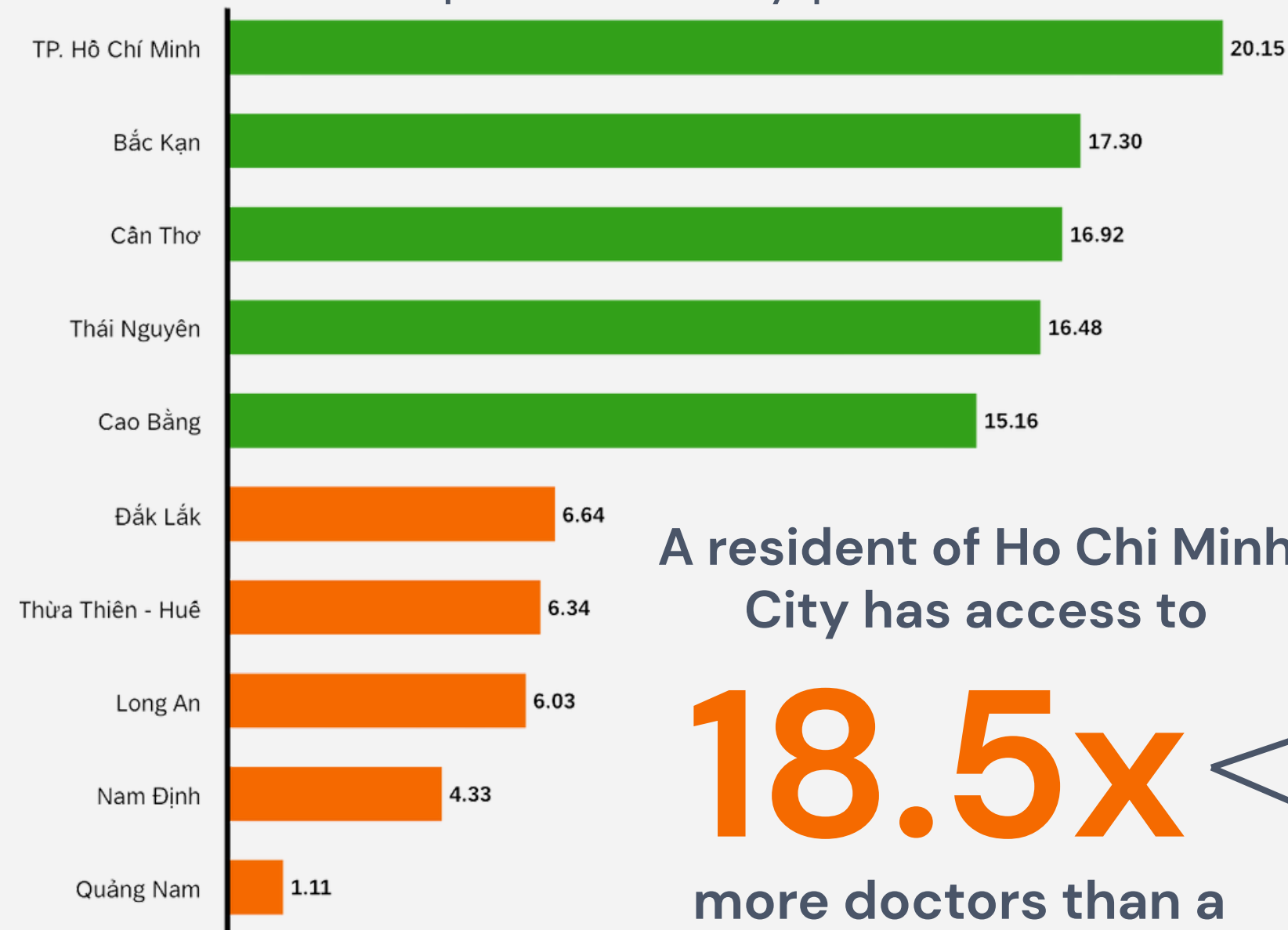
Source: Worldwide Cancer Research, theleader.vn

How UV Radiation Damages Skin at the Cellular Level



The Healthcare Gap – An 18-Fold Disparity

Medical doctors per 10000, by province



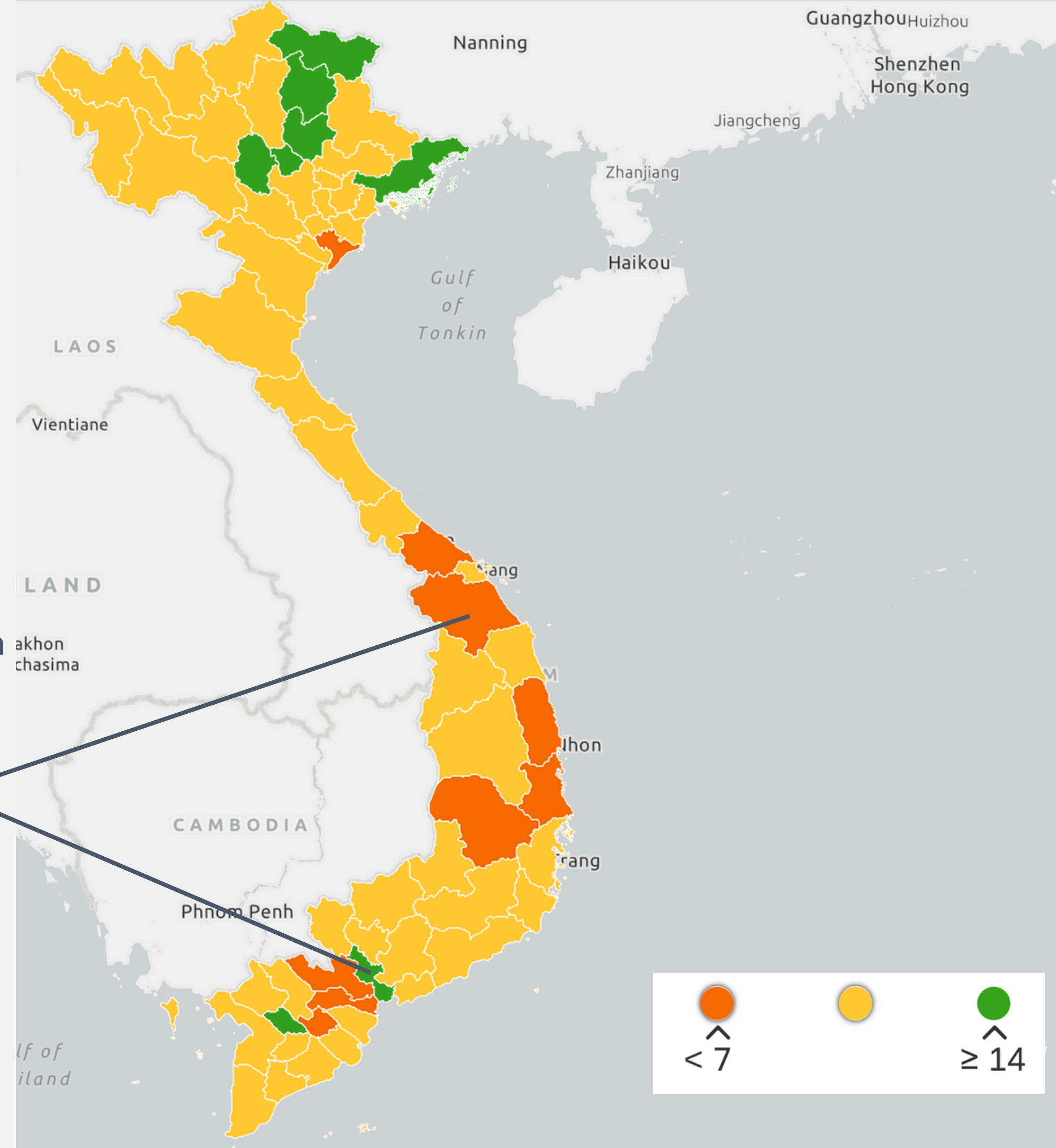
A resident of Ho Chi Minh
City has access to

18.5x

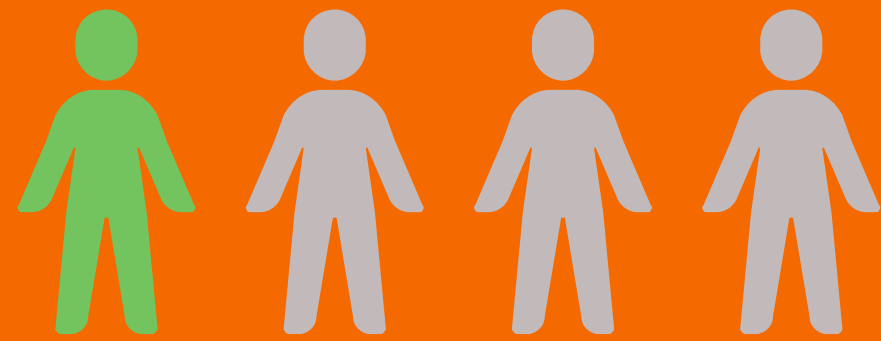
more doctors than a
resident in Quang Nam.

This isn't a gap; it's a chasm.

Source: General Statistics Office of Vietnam



The Stakes – Why Early Detection is Everything



1 in 4

Metastatic melanoma
patients survives
after 5 years of
discovery.

Survival rate of melanoma patients
after 5 years of discovery, by Stage

Localized melanoma (Stage 0, Stage I, and Stage II)



Regional melanoma (Stage III)



Metastatic melanoma (Stage IV)



This access gap has **life-or-death** consequences. Early, localized detection almost certainly results in survival. If diagnosis is delayed, that rate plummets by **75%**.

What the data has revealed



FIRST

a **harsher climate** is increasing the fundamental risk



SECOND

this disproportionately affects millions of essential **outdoor workers**



THIRD

a **severe, systemic gap** in healthcare access prevents this vulnerable population from getting timely screenings

This delay might turn a **highly survivable condition** into a **fatal one**.

The core problem we must solve is to overcome these problems to **eliminate the fatal delays**

Our Solution

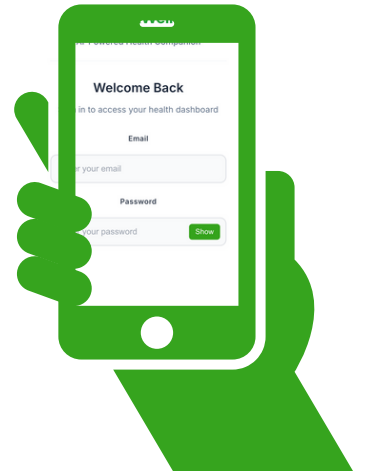


**SEEK
WELL**

A seamless ecosystem from screening to care, empowering community health with AI via democratized early detection



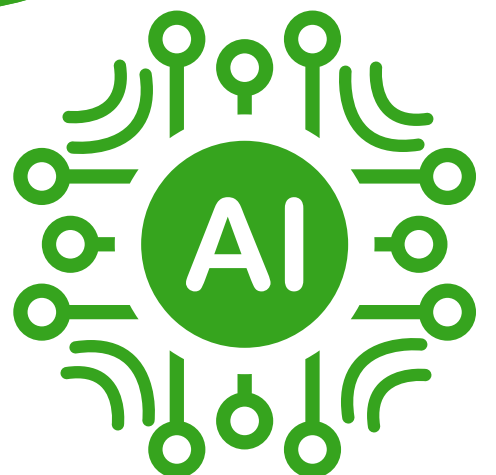
Local Patient



Local Cadre/
Smartphone



Doctor

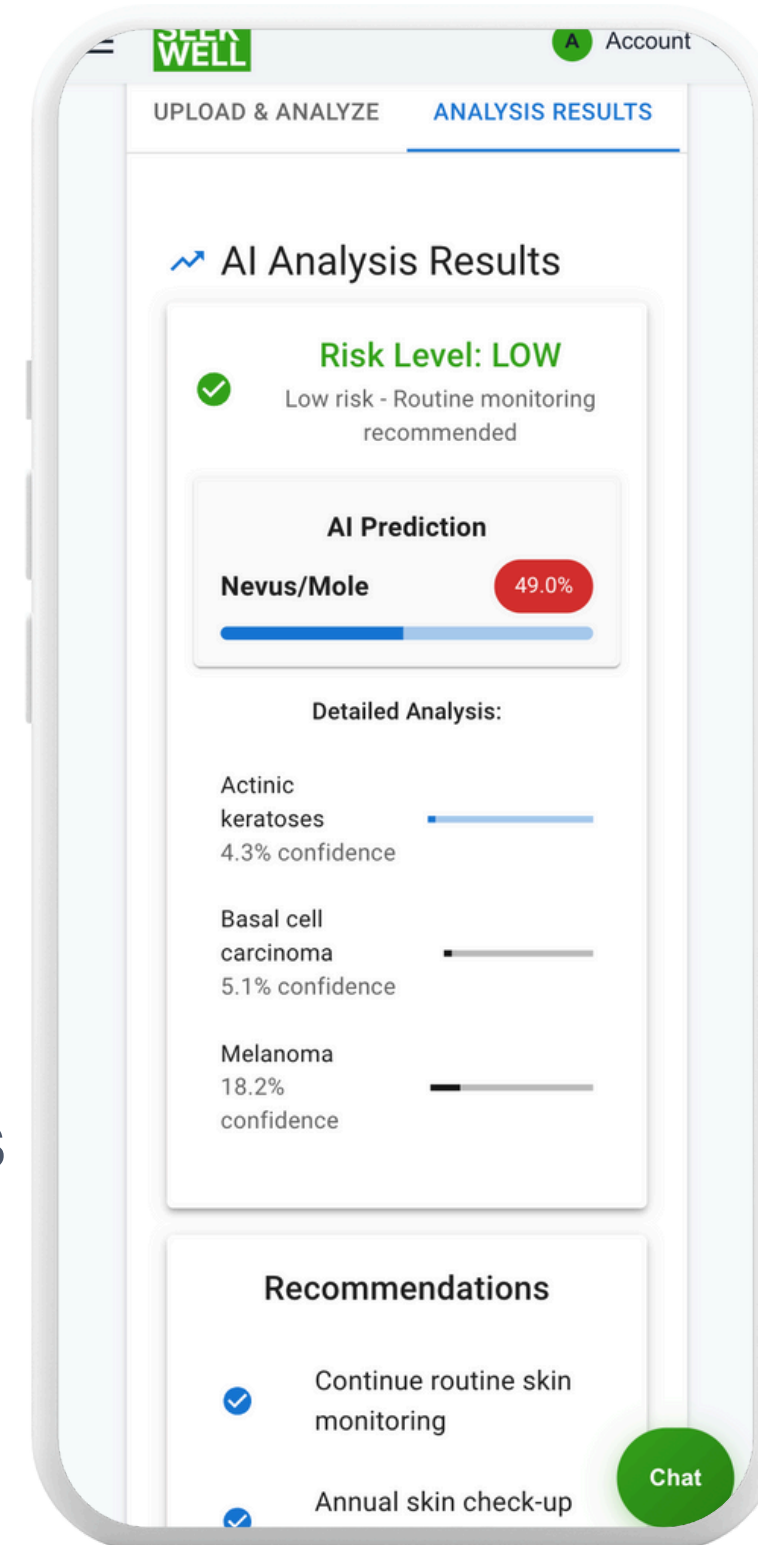


Advanced AI
System

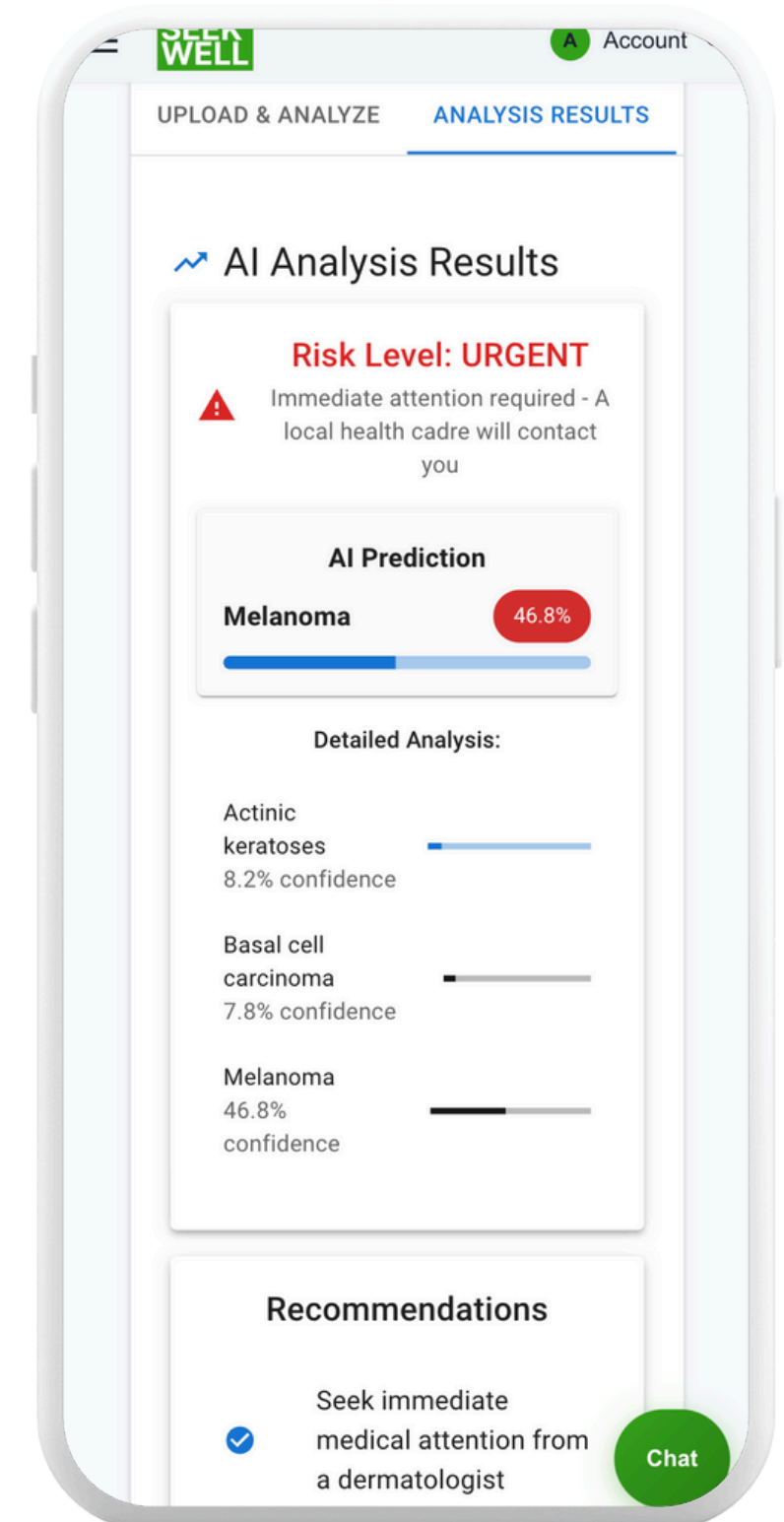
Technology in Action

SeekWell's State-of-the-art ViT AI Core

- **Robustly Trained:** On over **10,000 dermatoscopic images**, fine-tuned with **2,000 smartphones images** of various skin types, in different condition, to ensure equity and accuracy.
- **High Accuracy:** Delivers a risk classification with **70% accuracy**.
- **Immediate Results:** Provides analysis in **under 10 seconds** directly on a smartphone.



Reduces Healthcare Burden

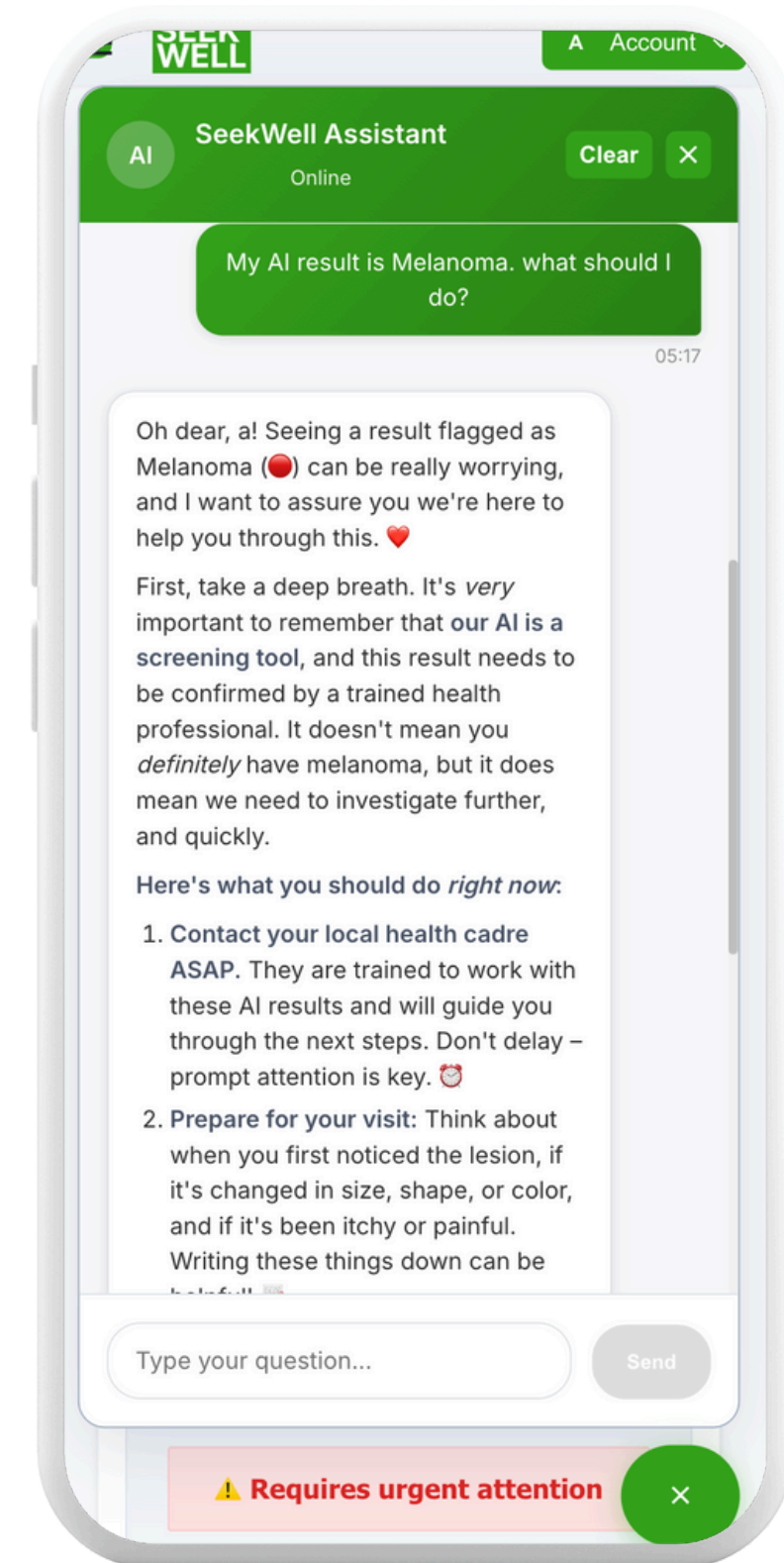


Enables Early Intervention

Technology in Action

Empathetic Guidance

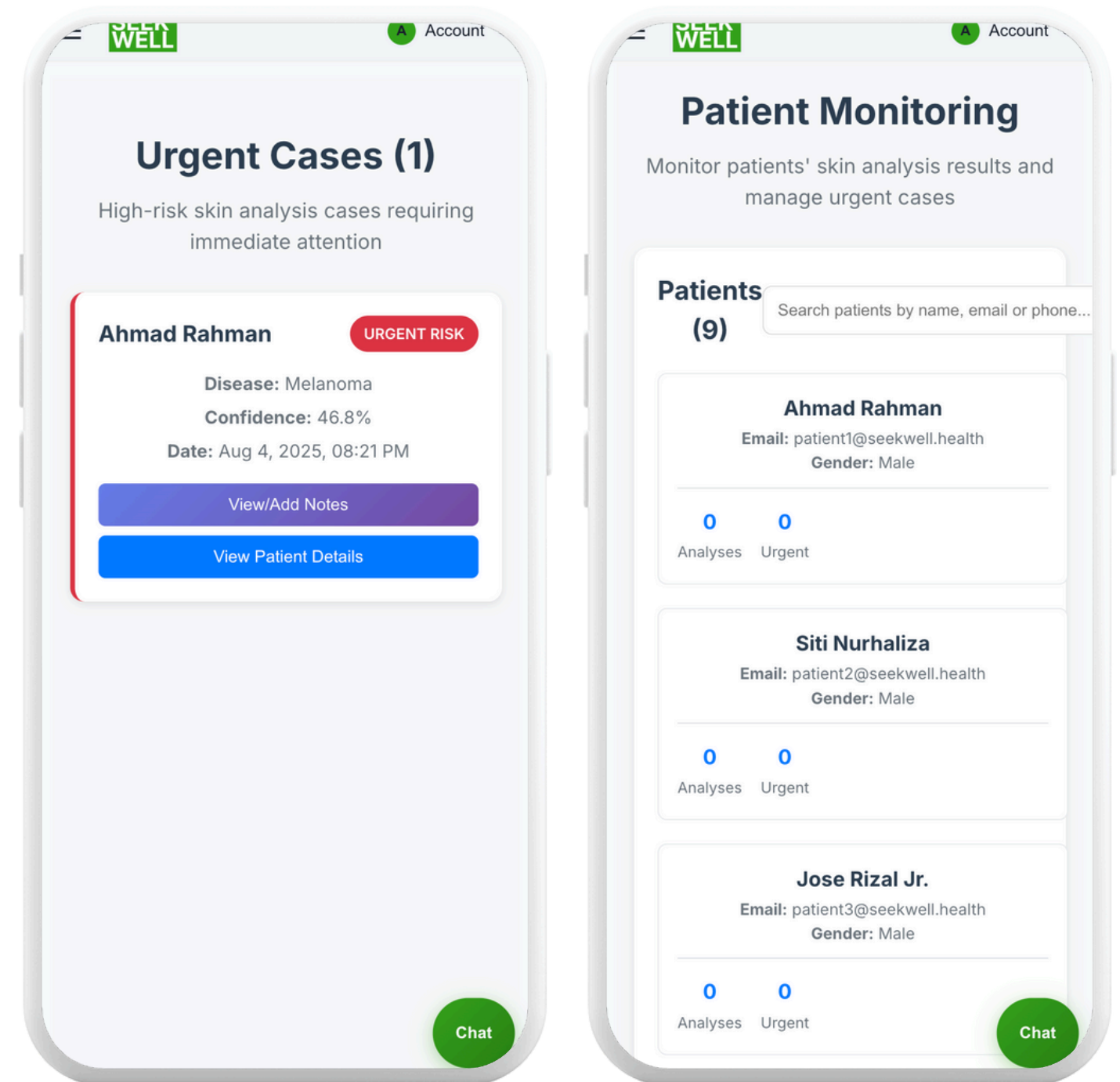
- **The Challenge:** An AI diagnosis can cause fear, leading to patient inaction.
- **Our Solution:** The SeekWell Assistant delivers results with empathy, empowering patients to move from anxiety to the crucial next step of getting care.
- **ASCC Alignment:** Builds an **Inclusive community with equitable access (B.2)** and a **Caring society (E.1)**.



Technology in Action

Empowering the Officials

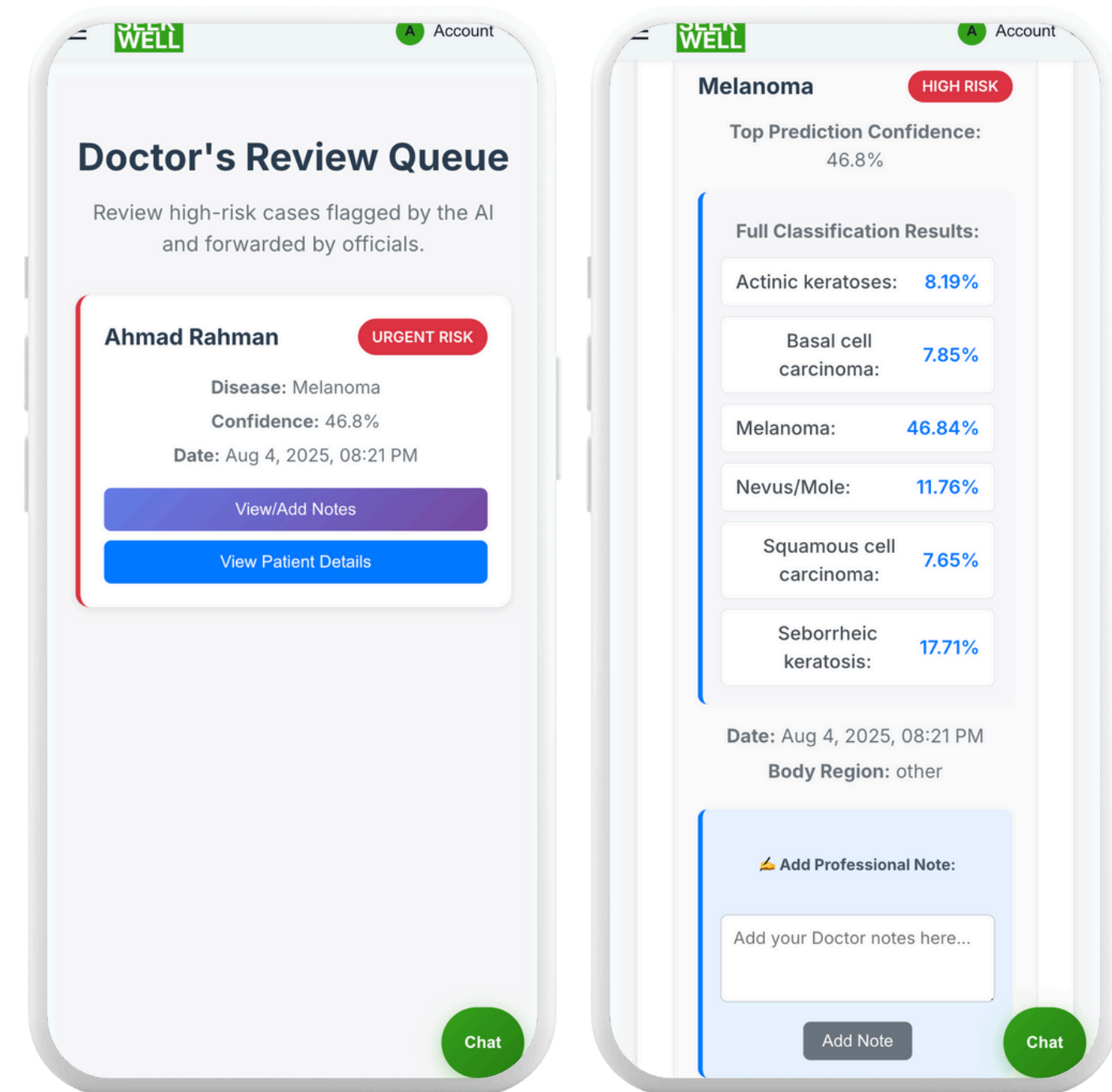
- **AI-Powered Triage:** Dashboard has a dedicated section prioritizing high-risk patients.
- **Official's Management:** Display detailed information of the community's analyses.
- **Impact:** Directly shortens the time-to-care, the most critical factor in improving skin cancer survival rates from 23% to 98%.



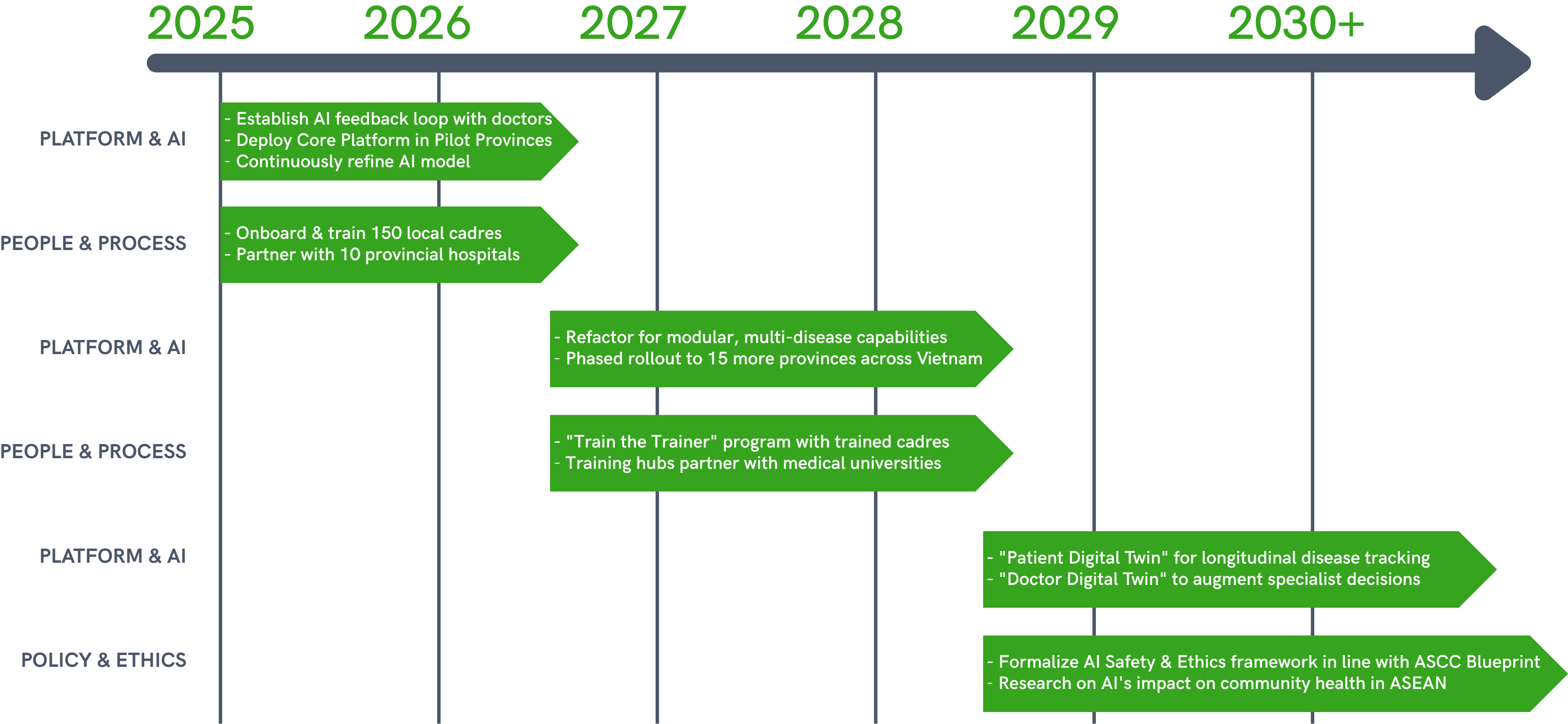
Technology in Action

Streamlining for the Specialist Doctor

- **Efficient Clinical Review:** Specialists receive a pre-filtered queue of the most urgent cases for immediate attention.
- **Data-Rich Referrals:** Each case includes the AI's assessment, confidence score, and notes from Patients and Officials for faster, informed decisions.



The SeekWell Roadmap



KPIs of Impact

10x INCREASED
ACCESS

To early screening for at-risk rural populations

60% REDUCTION

In unnecessary specialist referrals, easing healthcare system burden

90% REDUCTION

In the time from initial screening to professional assessment

2M+ PEOPLE
SERVED

Projected reach across ASEAN when the framework is scaled

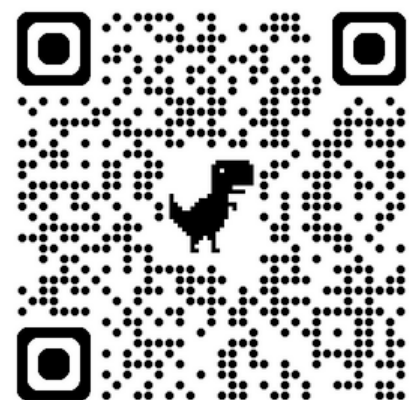
SDGs and ASCC Blueprint 2025 alignment

	Reduce skin cancer deaths through early detection.	Screening access for vulnerable rural outdoor workers.	Building a tech-driven resilient health system.
	TARGET 3-4  Reduce NCD deaths & promote mental health TARGET 3-8  Achieve Universal Health Coverage	TARGET 11-A  Strong National and Regional Development Planning	TARGET 13-1  Strengthen Resilience and Adaptive Capacity to Climate Related Disasters
	D.2 A Safer ASEAN that is able to Respond to all Health-related Hazards including Biological, Chemical, and Radiological-nuclear, and Emerging Threats	B.2 Equitable Access for All Equitable access to services, resources, and opportunities for all, especially the vulnerable and marginalised, to build an inclusive and resilient ASEAN.	D.4 Strengthened social protection for vulnerable and marginalised groups — including women, children, the elderly, persons with disabilities, ethnic minorities, migrants, and those in climate-sensitive or remote areas — to reduce risks during climate-related crises and disasters.

SEEK WELL

Turning early detection from a privilege into a right

seekwell.vercel.app



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